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Birds Image Classification Using Convolutional Neural Networks And VGGNet Architectures

Group Members

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1 Dataset Description

The dataset used in this experiment consists of bird images grouped into three classes: bird1, bird2, and bird3. The training dataset contained 3000 images while the testing dataset contained 900 images, with each class having 300 test samples. Before training, all images were resized to a resolution of 64×64 pixels and normalized to ensure stable model learning.

2 Methodology

2.1 Data Preprocessing

Image preprocessing included resizing all images to 64×64 pixels and normalizing pixel intensities to the range $[0, 1]$. Data augmentation techniques were applied during training to artificially increase dataset variability. These techniques included random rotations, horizontal flips, and small spatial shifts.

2.2 Regularization Techniques

To prevent overfitting, two regularization techniques were applied:

- **Dropout**: randomly deactivates neurons during training to improve generalization.
- **L2 Regularization**: penalizes large weight values to reduce model complexity.

2.3 Model Architectures

Two CNN architectures were implemented.

Custom CNN Architecture: The custom CNN consists of three convolutional layers followed by max-pooling operations and dropout layers. Feature maps extracted by the convolution layers are flattened and passed through a fully connected layer before classification using a softmax output layer.

VGG-style CNN Architecture: The second architecture follows the design principles of the VGG network, which uses stacked convolutional layers with small 3×3 filters. This architecture contains multiple convolutional blocks followed by pooling layers and fully connected layers for classification.

3 Experimental Results

Both models were trained for 20 epochs using the Adam optimizer and sparse categorical cross-entropy as the loss function.

3.1 Model Performance

Table 1 summarizes the overall performance of the two models.

Table 1: Overall Model Performance

Model	Loss	Accuracy	Observations
Custom CNN	0.5829	0.8133	Lower loss and slightly better performance
VGGNet	0.7132	0.8122	Comparable accuracy but slightly higher loss

3.2 Class-wise Performance

Table 2 presents the precision, recall, and F1-score for each bird category using the Custom CNN model.

Table 2: Class-wise Performance (Custom CNN)

Class	Precision	Recall	F1-score
Bird1	0.97	0.84	0.90
Bird2	0.72	0.81	0.77
Bird3	0.78	0.79	0.78

The results indicate that bird1 images were the easiest to classify, achieving the highest precision and F1-score. Bird2 and bird3 exhibited slightly lower performance, suggesting that these categories may share similar visual characteristics.

4 Training and Evaluation Visualizations

To better understand the training behavior and prediction performance, four visualizations were generated.

4.1 Training History Plots

The training history plots show the evolution of training and validation accuracy and loss across epochs.

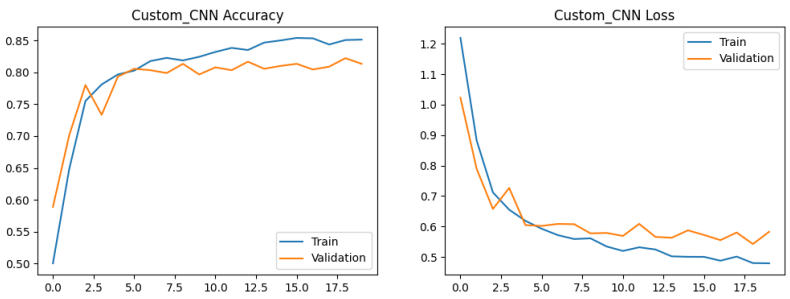

Figure 1: Training and Validation Performance of Custom CNN

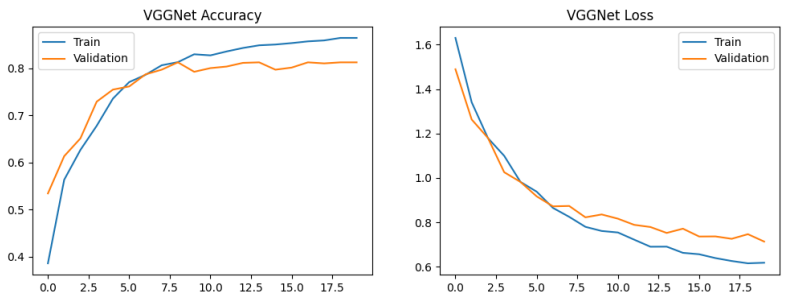

Figure 2: Training and Validation Performance of VGGNet

4.2 Confusion Matrix Analysis

Confusion matrices provide insights into the types of classification errors made by each model.

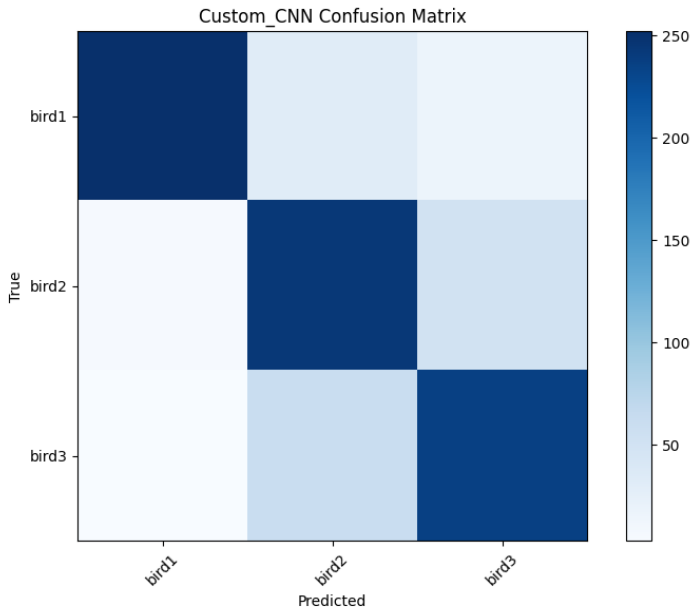

Figure 3: Confusion Matrix for Custom CNN

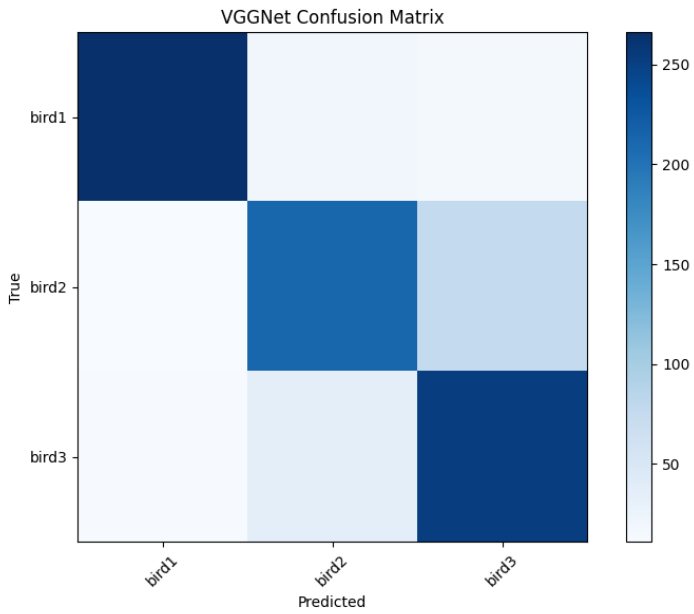

Figure 4: Confusion Matrix for VGGNet

5 Comparison of the Results

Table 3: Performance Comparison Between Custom CNN and VGGNet

Metric	**Custom CNN**	**VGGNet**	**Best**
Loss	0.5829	0.7132	Custom CNN
Accuracy	0.8133	0.8122	Custom CNN
Bird1 F1	0.90	0.90	Tie
Bird2 F1	0.77	0.75	Custom CNN
Bird3 F1	0.78	0.78	Tie
Macro F1	0.82	0.81	Custom CNN

6 Conclusion

This project successfully implemented and evaluated convolutional neural networks for bird image classification. Two architectures were compared: a custom-designed CNN and a simplified VGG-style CNN.

Both models achieved comparable accuracy levels, with the custom CNN slightly outperforming the VGG architecture in terms of loss and overall performance. The experiment demonstrates that relatively simple CNN architectures can achieve strong classification performance when supported by appropriate preprocessing, data augmentation, and regularization strategies.